

The equations $2^N \pm 2^M \pm 2^L = z^{2*}$

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1. INTRODUCTION

In the present paper we solve the title equations. It is easy to see that they lead either to

$$(1) \quad 2^n \pm 2^m \pm 1 = x^2,$$

or to

$$(2) \quad 2^n \pm 2^m \pm 2 = x^2.$$

While the examination of (2) is quite simple, as well as the resolution of $2^n \pm 2^m - 1 = x^2$, the equation

$$(3) \quad 2^n + 2^m + 1 = x^2$$

requires more calculations and the application of some deep results of Beukers [2]. This problem has been posed by professor Tijdeman, and I heard it from Tengely.

From a wider point of view, equations of types similar to (1) and (2) have already been investigated. Gerono [4] proved that a Mersenne-number $M_k = 2^k - 1$ cannot be a power of a natural number if $k > 1$, so the equation

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$2^k - 1 = x^2$ has only the solutions $(k, x) = (0, 0), (1, 1)$. For another example, it can readily be verified that $2^k + 1 = x^2$ implies $(k, x) = (3, 2)$.

Ramanujan [7] conjectured that the diophantine equation

$$(4) \quad 2^k - 7 = x^2$$

has five solutions, namely $(k, x) = (3, 1), (4, 3), (5, 5), (7, 11)$ and $(15, 181)$. His conjecture was first proved by Nagell [6]. The generalized Ramanujan-Nagell equation

$$(5) \quad 2^k + D = x^2$$

in natural numbers k and x , where $D \neq 0$ is an integer parameter, was considered by several authors. See, for example, Apéry [1], Hasse [5], Beukers [2]. Taking $D = \pm 2^M \pm 2^L$, we investigate infinitely many generalized Ramanujan-Nagell equations.

Our main result is Theorem 1. Theorems 1 and 2 have interesting consequences connected to binary recurrences (Corollary 1). Finally, a corollary of Lemma 5 states that the ratio of two distinct triangular numbers cannot be a power of 4 (Corollary 2).

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2. RESULTS

Theorem 1. *If the positive integers n , m and x with $n \geq m$ satisfy*

$$(6) \quad 2^n + 2^m + 1 = x^2,$$

then

- (i) $(n, m, x) \in \{(2t, t + 1, 2^t + 1) \mid t \in \mathbb{N}, t \geq 1\}$ or
- (ii) $(n, m, x) \in \{(5, 4, 7), (9, 4, 23)\}$.

Remarks. I. Equation (6), essentially, asks for odd natural numbers x whose squares contain exactly three 1 digits with respect to the base 2. Theorem 1 says that beside the infinite set $x^2 = 101_2^2 = 11001_2$, $1001_2^2 = 1010001_2$, $\dots$, only $x^2 = 111_2^2 = 110001_2$ and $x^2 = 10111_2^2 = 1000010001_2$ possess the property above.

II. The solutions (i) and (ii) in Theorem 1 enable to determine all $n, m \in \mathbb{Z}$, $x \in \mathbb{Q}$ satisfying (6).

Theorem 2. *If the positive integers n , m and x satisfy*

$$(7) \quad 2^n - 2^m + 1 = x^2,$$

then

- (i) $(n, m, x) \in \{(2t, t + 1, 2^t - 1) \mid t \in \mathbb{N}, t \geq 2\}$ or

- (ii) $(n, m, x) \in \{(t, t, 1) \mid t \in \mathbb{N}, t \geq 1\}$ or
- (iii) $(n, m, x) \in \{(5, 3, 5), (7, 3, 11), (15, 3, 181)\}$.

Theorem 3. *If the positive integers n, m and x with $n \geq m$ satisfy*

$$(8) \quad 2^n + 2^m - 1 = x^2,$$

then

$$(i) \quad (n, m, x) = (3, 1, 3).$$

Moreover, all the solutions of the equation

$$(9) \quad 2^n - 2^m - 1 = x^2$$

in positive integers n, m and x are given by

$$(ii) \quad (n, m, x) = (2, 1, 1).$$

One can find lots of results concerning occurrence of squares and higher powers in binary (or higher order) recurrences. See, for instance, Shorey, Tijdeman [8], Chapter 9. Corollary 1 determines all square terms in certain binary recursive sequences.

Corollary 1. *(Corollary of Theorems 1 and 2.) Let d be an arbitrarily fixed natural number. Consider the binary recurrences*

$$(10) \quad G_m = 3G_{m-1} - 2G_{m-2} \quad (m \geq 2), \quad G_0 = 2^d + 2, \quad G_1 = 2^{d+1} + 3;$$

$$(11) \quad H_m = 3H_{m-1} - 2H_{m-2} \quad (m \geq 2), \quad H_0 = 2^d, \quad H_1 = 2^{d+1} - 1.$$

(i) The only square occurring in the recursive sequence G is G_{d+2} , except for the following two cases. If $d = 1$, then G contains three squares, namely $G_0, G_3 = G_{d+2}$ and G_4 . If $d = 5$, then G_4 and $G_7 = G_{d+2}$ are the squares in G .

(ii) If d is odd, then

$$(12) \quad H_m = w^2$$

implies $m = d + 2$. If $d > 0$ is even, then equation (12) has exactly two solutions given by $m = 0$ and $m = d + 2$, except for three cases $d = 2, 4, 12$ when there is an additional square, viz. H_3 .

The second corollary contributes to the colorful palette of the results concerning triangular numbers.

Corollary 2. *(Corollary of Lemma 5.) Let $\triangle_k$ denote the k^{th} triangular number, i.e. $\triangle_k = \frac{k(k+1)}{2}$, $k \geq 1$, $k \in \mathbb{N}$. Then the diophantine equation*

$$(13) \quad \frac{\triangle_y}{\triangle_x} = 4^t, \quad y \neq x$$

has no solution in natural numbers x, y and t .

3. PRELIMINARIES

Lemma 1. *Let $D_1 \in \mathbb{Z}$, $D_1 \neq 0$. If $|D_1| < 2^{96}$ and $2^n + D_1 = x^2$ has a solution (n, x) then*

$$(14) \quad n < 18 + 2 \log_2 |D_1|.$$

Proof. This is Corollary 2 in [2] due to Beukers.

Lemma 2. *Let p be an odd power of 2. Then for all $x \in \mathbb{Z}$*

$$(15) \quad \left| \frac{x}{p^{0.5}} - 1 \right| > \frac{2^{-43.5}}{p^{0.9}}.$$

Proof. We refer again to Beukers, [2].

Lemma 3. *Let $D_2 \in \mathbb{N}$ be odd. The equation $2^n - D_2 = x^2$ has two or more solutions in positive integers n, x if and only if $D_2 = 7, 23$ or $2^k - 1$ for some $k \geq 4$. The solutions, in these exceptional cases, are given by the following table.*

$D_2 = 7$	$(n, x) = (3, 1), (4, 3), (5, 5), (7, 11), (15, 181),$
$D_2 = 23$	$(n, x) = (5, 3), (11, 45),$
$D_2 = 2^k - 1, (k \geq 4)$	$(n, x) = (k, 1), (2k - 2, 2^{k-1} - 1).$

Proof. See Theorem 2 in [2].

Lemma 4. *All natural solutions (n, x) of the inequalities*

$$(16) \quad 0 < |2^n - x^2| < 4$$

in positive integers n and x are given by $(n, x) \in \{(1, 1), (2, 1), (3, 3), (1, 2)\}$.

Proof. In virtue of Lemma 1, $n < 22$ and the verification of all possible values n gives the solutions above. $\square$

Lemma 5. *Let t be an arbitrary positive integer. If x and y are integers satisfying*

$$(17) \quad y^2 - 1 = 2^{2t}(x^2 - 1), \quad y > 1, \quad x > 1,$$

then $x = 2^{t-1}$ and $y = 2^{2t-1} - 1$ for $t > 1$.

Proof. It is easy to see that (17) is not solvable if $t = 1$. Suppose that $t > 1$, $y > 1$ and $x > 1$ satisfy (17). Then y is odd and

$$(18) \quad \frac{y-1}{2} \cdot \frac{y+1}{2} = 2^{2t-2}(x^2 - 1).$$

The greatest common divisor of $\frac{y-1}{2}$ and $\frac{y+1}{2}$ is 1 and $2^{2t-2} (\geq 4)$ divides exactly one of the terms on the left hand side of (18). Consequently, $y = 2^{2t-1}k \pm 1$

with some integer $k \geq 1$. By (17) we have $y < 2^t x$, therefore $2^{t-1}k \leq x$. Moreover, it follows that

$$(19) \quad 2^{2t-2}k^2 - k = x^2 - 1 \quad \text{or} \quad 2^{2t-2}k^2 + k = x^2 - 1.$$

In the first case, clearly, $k = 1$ provides the solution $x = 2^{t-1}$, $y = 2^{2t-1} - 1$. If $k > 1$, then the inequalities

$$(20) \quad x^2 = 2^{2t-2}k^2 - (k-1) < 2^{2t-2}k^2 \leq x^2$$

lead to contradiction.

In the second case of (19) it follows that

$$(21) \quad (2^{t-1}k)^2 < (2^{t-1}k)^2 + k + 1 = x^2 < (2^{t-1}k + 1)^2,$$

which is impossible. $\square$

Lemma 6. *Let n, m and x be positive integers satisfying $2 \leq m < n$ and*

$$(22) \quad 2^n + 2^m + 1 = x^2.$$

Then $x = 2^{m-l}(2k+1) \pm 1$ with some $k \in \mathbb{N}$.

Proof. Assume that (n, m, x) is a solution of (22) under the assumptions made. For $m = 2$ the lemma trivially states that x is odd. If $m \geq 3$, then the congruence

$$(23) \quad x^2 \equiv 1 \pmod{2^m}$$

has exactly four incongruent solutions, namely $x \equiv 1$, $x \equiv 2^{m-1} - 1$, $x \equiv 2^{m-1} + 1$ and $x \equiv 2^m - 1 \pmod{2^m}$.

The first and fourth cases are impossible because, by (22), $x = 2^m l \pm 1$, ($l \in \mathbb{N}$, $l \geq 1$) leads to

$$(24) \quad 2^{n-m} + 1 = 2^m l^2 \pm 2l.$$

The second and third solutions of (23) provide

$$(25) \quad x = 2^{m-1}(2k+1) \pm 1, \quad (k \in \mathbb{N}). \quad \square$$

Lemma 7. *If n, m and x are natural numbers for which $m < n$ and $n < 2m - 2$, then*

$$(26) \quad 2^n + 2^m + 1 = x^2$$

implies $(n, m, x) = (5, 4, 7)$.

Proof. The conditions of the lemma give $m \geq 4$. Suppose that (n, m, x) satisfy (26). Combining Lemma 6 and (26) we obtain

$$(27) \quad 2^n + 2^m + 1 = r^2 2^{2m-2} \pm r 2^m + 1,$$

where r is a positive odd integer. Since $2m - 2 \geq n + 1$, we get

$$(28) \quad 2^m(1 \mp r) \geq (2r^2 - 1)2^n.$$

Hence $r = 1$, $x = 2^{m-1} - 1$ and $n \leq m + 1$. By $m < n$ we have $n = m + 1$ and we can conclude that $m = 4$, $n = 5$ and $x = 7$.

Lemma 8. *If D , k and x are positive integers, $k \geq 3$ and*

$$(29) \quad 2^{D+8k} + 2^{4k} + 1 = x^2,$$

then $D > 56k - 32$.

Proof. Let $\nu_2(n)$ denote the 2-adic value of the integer n . Assume that the integers D , k and x satisfy the conditions of the lemma. By Lemma 6 we have two possibilities for x .

A) First consider the case $x = 2^{4k-1}(2u_0 + 1) + 1$, ($u_0 \geq 0$). By (29) we obtain

$$(30) \quad 2^{D+4k-1} = 2^{4k-3}(2u_0 + 1)^2 + u_0,$$

where u_0 must be positive and $u_0 = 2^{4k-3}u_1$ with some positive odd integer u_1 . Otherwise, dividing (30) by $2^{\min\{4k-3, \nu_2(u_0)\}}$, it leads to contradiction. In the sequel, this type of argument will be applied without any further notice. It follows that

$$(31) \quad 2^{D+2} = 2^{8k-4}u_1^2 + 2^{4k-1}u_1 + (u_1 + 1).$$

Then $u_1 + 1 = 2^{4k-1}u_2$ for some suitable positive odd integer u_2 , and by (31) we get

$$(32) \quad 2^{D-4k+3} = 2^{4k-3}(2^{4k-1}u_2 - 1)^2 + 2^{4k-1}u_2 + (u_2 - 1).$$

Clearly, $u_2 \neq 1$, $u_2 - 1 = 2^{4k-3}u_3$, ($u_3 \in \mathbb{N}$, $u_3 \equiv 1 \pmod{2}$), further

$$(33) \quad \begin{aligned} 2^{D-8k+6} &= 2^{8k-2}(2^{4k-3}u_3 + 1)^2 - 2^{4k}(2^{4k-3}u_3 + 1) \\ &\quad + 2^{4k-1}u_3 + (u_3 + 5). \end{aligned}$$

It is easy to see that $u_3 + 5 = 2^{4k-1}u_4$, where u_4 is an odd natural number. Hence

$$(34) \quad \begin{aligned} 2^{D-12k+7} &= 2^{4k-1}(2^{4k-3}(2^{4k-1}u_4 - 5) + 1)^2 - \\ &\quad - 2^{4k-2}(2^{4k-1}u_4 - 5) + 2^{4k-1}u_4 + (u_4 - 7). \end{aligned}$$

By (34) we conclude that $u_4 - 7 = 2^{4k-2}u_5$. Here the odd integer $u_5 = \frac{u_4-7}{2^{4k-2}}$ is positive because $k \geq 3$ and $u_4 > 0$. It follows that

$$(35) \quad \begin{aligned} 2^{D-16k+9} &= 2^{8k-5}(2^{4k-1}(2^{4k-2}u_5 + 7) - 5)^2 + 2^{8k-2}(2^{4k-2}u_5 + 7) - \\ &\quad - 2^{8k-3}u_5 + (u_5 - 12)2^{4k-1} + (u_5 + 21). \end{aligned}$$

Finally, $u_5 + 21 = 2^{4k-1}u_6$, ($u_6 \in \mathbb{N}$, $u_6 \equiv 1 \pmod{2}$), and then $u_6 - 33 = 2^{4k-4}u_7$ leads to the equality

$$(36) \quad 2^{D-24k+14} = (2^{4k-1}(2^{4k-2}Q_1 + 7) - 5)^2 + R_1 + S_1,$$

where

$$(37) \quad Q_1 = 2^{4k-1}(2^{4k-4}u_7 + 33) - 21, \quad R_1 = 2^3(2^{4k-2}Q_1 + 7) - 2^2Q_1,$$

$$(38) \quad S_1 = 2^3(2^{4k-4}u_7 + 33) + u_7, \quad u_7 \in \mathbb{N}, \quad u_7 \equiv 1 \pmod{2}.$$

Obviously, $Q_1 > 2^{8k-5}$, $S_1 > 0$, $R_1 > 0$. Therefore $2^{D-24k+14} > 2^{32k-16}$ and

$$(39) \quad D > 56k - 30.$$

B) In the second case replace x by $2^{4k-1}(2u_0 + 1) - 1$, ($u_0 \geq 0$) in (29) and, similarly as above, the substitutions $u_0 = 2^{4k-3}u_1 - 1$, $u_1 = 2^{4k-1}u_2 + 1$, $u_2 = 2^{4k-3}u_3 - 1$, $u_3 = 2^{4k-1}u_4 + 5$, $u_4 = 2^{4k-2}u_5 - 7$, $u_5 = 2^{4k-1}u_6 + 21$ and $u_6 = 2^{4k-4}u_7 - 33$ lead to the equality

$$(40) \quad 2^{D-24k+14} = (2^{4k-1}Q_2 + 5)^2 - 8Q_2 + R_2,$$

where

$$(41) \quad Q_2 = 2^{4k-2}(2^{4k-1}(2^{4k-4}u_7 - 33) + 21) - 7,$$

$$(42) \quad R_1 = 2^2(2^{4k-1}(2^{4k-4}u_7 - 33) + 21) - 2^3(2^{4k-4}u_7 - 33) - u_7,$$

and $u_7 \in \mathbb{N}$, $u_7 \equiv 1 \pmod{2}$. It can be proved that $Q_2 > 2^{12k-8}$, $R_2 > 0$ and we have $2^{D-24k+14} > 2^{32k-18}$, which, together with (39), implies $D > 56k - 32$.

The proof of Lemma 8 is complete. $\square$

Lemma 9. *If a and c are non-negative integers satisfying*

$$(43) \quad a^2 + (a+1)^2 = c^2,$$

then $a = 2P_nP_{n+1}$, $a+1 = P_{n+1}^2 - P_n^2$ or conversely, where P_k denotes the k^{th} term of the Pell sequence defined by $P_0 = 0$, $P_1 = 1$ and $P_k = 2P_{k-1} + P_{k-2}$, ($k \geq 2$).

Proof. Probably this is an old result. For the proof see, for instance, Cohn [3].

Proof of Theorem 1. Obviously, each element of the set

$$(44) \quad T = \{(n, m) \in \mathbb{N}^2 \mid n = 2t, m = t + 1, t \in \mathbb{N}, t \geq 1\}$$

(with some suitable $x \in \mathbb{N}$) satisfies the relations

$$(45) \quad 2^n + 2^m + 1 = x^2, \quad n \geq m \geq 1.$$

Let S denote the set of solutions (n, m) of (45), further let $M_1 = (5, 4)$ and $M_2 = (9, 4)$. We have to show that the set of exceptional solutions is $S \setminus T = \{M_1, M_2\}$.

Observe that $2^{n-m} + 1 = \frac{x^2 - 1}{2^m} \in \mathbb{N}$ if $(n, m) \in S$, further

$$(46) \quad 2^{2n-2m} + 2^{n-m+1} + 1 = \left(\frac{x^2 - 1}{2^m} \right)^2.$$

Hence a solution (n, m) of (45) provides $(2n - 2m, n - m + 1) \in S$, except when $2n - 2m < n - m + 1$, i.e. $n = m$. But Lemma 4 implies that the only solution (n, m) with $n = m$ is $(2, 2) \in T$.

In the sequel, we assume that $n > m$. Then the transformation

$$(47) \quad \tau : (n, m) \mapsto (2n - 2m, n - m + 1), \quad (n > m)$$

induces a map of $S \setminus \{(2, 2)\}$ into S .

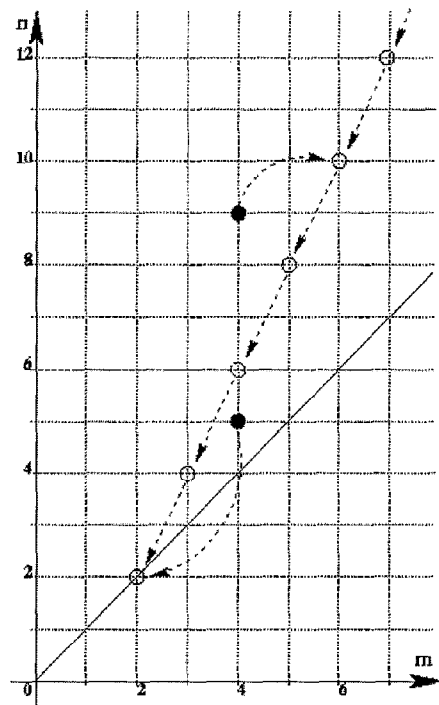


Figure 1. Map τ on the solutions of the equation $2^n + 2^m + 1 = x^2$.

The map τ has important properties. If $(n, m) \in S$, then let $\delta(n, m)$ denote the distance $n - m$ of the exponents n and m .

Property 1. $\delta(\tau(n, m)) = \delta(n, m) - 1$. In particular, $\tau(n, m) \neq (n, m)$, i.e. the map has no fixed points.

Property 2. If $(n, m) \in T \setminus \{(2, 2)\}$, more precisely if $(n, m) = (2t, t + 1)$, $t \geq 2$, then $\tau(n, m) = (2(t - 1), t) \in T$ is the ‘lower neighbour’ solution of (n, m) in T . Thus the elements of the set T are ordered by τ . Moreover $\delta(\tau(2t, t + 1)) = t - 2$, ($t \geq 2$) shows that all natural numbers occur as a difference of the exponents in the solution of (45).

Property 3. If (n, m) is an exceptional solution (i.e. $(n, m) \in S \setminus T$), then $\tau(n, m) \in T$ since $\tau(n, m) = (2\delta(n, m), \delta(n, m) + 1)$. Especially, $\tau(5, 4) = (2, 2)$, $\tau(9, 4) = (10, 6)$.

If $m = 1$, then Lemma 4 implies a solution with $n < m$, which contradicts the assertion $n > m$.

Now suppose that the integers n and m satisfy $2 \leq m < n$ and (45). Reconsidering the map

$$(48) \quad \tau : S \setminus \{(2, 2)\} \mapsto S$$

with (47), by Properties 1-3 we have to prove that there are exactly two cases when $(n, m) \neq (n_1, m_1)$ and $\tau(n, m) = \tau(n_1, m_1)$. In other words, we must show that the system of the equations

$$(49) \quad 2^n + 2^m + 1 = x^2$$

$$(50) \quad 2^{n+d} + 2^{m+d} + 1 = y^2$$

in positive integers n, m, d, x, y with $2 \leq m < n$ has exactly two solutions.

Taking such a solution, obviously both $x > 1$ and $y > 1$ are odd. It follows from (49) and (50) that

$$(51) \quad y^2 - 1 = 2^d(x^2 - 1),$$

and by Lemma 5 we infer that d must be odd.

Observe that one of (n, m) and $(n + d, m + d)$ has to belong to the set $T \setminus \{(2, 2)\}$. On the contrary, if both (n, m) and $(n + d, m + d)$ are exceptional, by the properties of the transformation τ there exists a solution $(n_2, m_2) \in T \setminus \{(2, 2)\}$ such that $\tau(n_2, m_2) = \tau(n, m) = \tau(n + d, m + d)$. But in this case one of the distances $|n_2 - n| = |m_2 - m|$ and $|n_2 - (n + d)| = |m_2 - (m + d)|$ has to be even since d is odd, which contradicts again to Lemma 5. Therefore, we distinguish two cases.

A) First let (50) be the exceptional case, consequently $(n, m) \in T \setminus \{(2, 2)\}$, and by (44) it follows that $n = 2m - 2$, which, together with (50), implies

$$(52) \quad 2^{2m-2+d} + 2^{m+d} + 1 = y^2.$$

Here, if $m \geq 3$, then the exponents $m + d$ and $2m - 2 + d$ on the left hand side satisfy the conditions of Lemma 7. Thus we conclude that $m = 3, d = 1, y = 7$ is the only solution of (52) (and $n = 4, x = 5$ of (49)). It gives $M_1 \in S$. On the other hand, if $m = 2$, then $n = 2m - 2 \leq m$ leads to contradiction.

B) The second possibility is that (49) is the exceptional case, while $(n + d, m + d) \in T \setminus \{(2, 2)\}$, i.e. $n = 2m + d - 2$. Then by (49) we have

$$(53) \quad 2^{2m+d-2} + 2^m + 1 = x^2.$$

It is easy to show that one of the exponents must be even in (53). Since d is odd, therefore m has to be even. Put $m = 2r$, where $r \in \mathbb{N}, r \geq 1$, and let $D = d - 2$. If $D = -1$, then (53) is equivalent to

$$(54) \quad 2^{4r-1} + 2^{2r} + 1 = x^2.$$

Observe that the left hand side of (54) is a sum of $(2^{2r-1})^2$ and $(2^{2r-1} + 1)^2$, hence $a = 2^{2r-1}$, $a + 1$ and x form a Pythagorean triple. Since a is even, by Lemma 9 we have $2^{2r-1} = 2P_nP_{n+1}$ with some $n \in \mathbb{N}$. Therefore, both P_n and P_{n+1} are power of 2, which is impossible if $n \geq 2$ because P_n and P_{n+1} are co-prime. Since $P_0 = 0$, the only possibility is $n = 1$, but $1 \cdot 2 \neq 2^{2r-2}$.

Consequently, $D \geq 1$ and we have

$$(55) \quad 2^{D+4r} + 2^{2r} + 1 = x^2.$$

The left hand side of (55) is quadratic residue (mod 5) if and only if r is even. Put $r = 2k$, ($k \in \mathbb{N}$, $k \geq 1$). Thus

$$(56) \quad 2^{D+8k} + 2^{4k} + 1 = x^2,$$

which is equivalent to

$$(57) \quad \frac{x}{2^{\frac{D+8k}{2}}} - 1 = \frac{2^{4k} + 1}{2^{\frac{D+8k}{2}} \left(x + 2^{\frac{D+8k}{2}} \right)}.$$

Applying Lemma 2 to the left hand side of (57), and using that (56) gives $2^{\frac{D+8k}{2}} < x$, we obtain

$$(58) \quad \frac{2^{-43.5}}{2^{(D+8k) \cdot 0.9}} < \frac{2^{4k} + 1}{2 \cdot 2^{D+8k}}.$$

We see that $2^{4k} + 1 < 2^{4k+0.5}$ if $k \geq 1$, and by (58) it follows that

$$(59) \quad D < 32k + 430.$$

On the other hand, considering (56), Lemma 8 provides $D > 56k - 32$, which, together with (59) implies $k \leq 19$. Finally, applying Lemma 1 to (56) with $D_1 = 2^{4k} + 1$, ($k \leq 19$) we conclude that $D \leq 19$, too. A simple computer search shows that equation (56) with odd $D \leq 19$ and $k \leq 19$ has only one solution $D = 1$, $k = 1$, $x = 23$. Hence we obtain the third exceptional solution of (45): $(n, m) = (D + 8k, 4k) = (9, 4)$, and there are no others. So the proof of Theorem 1 is complete.

Proof of Theorem 2. Suppose that $(n, m, x) \in \mathbb{N}^3$ is a positive solution of the diophantine equation

$$(60) \quad 2^n - 2^m + 1 = x^2.$$

Consider the case $n \geq m$. First let $m \geq 4$. Then (60) is equivalent to the equation

$$(61) \quad 2^n - D_2 = x^2,$$

where the positive number $D_2 = 2^m - 1$ is odd. By Lemma 3, we find

$$(62) \quad (n, x) = (m, 1), (2m - 2, 2^{m-1} - 1)$$

as the set of all the solutions of (61) with $m \geq 4$. This result leads to the following solutions of (60):

$$(63) \quad (n, m, x) = (t, t, 1), \quad t \in \mathbb{N}, \quad t \geq 4;$$

$$(64) \quad (n, m, x) = (2t, t + 1, 2^t - 1), \quad t \in \mathbb{N}, \quad t \geq 3.$$

The famous case $m = 3$ of (60) has five solutions given by the table in Lemma 3. Among them $(n, m, x) = (3, 3, 1)$ can be joined to the set (63) with the parameter $t = 3$, moreover, $(n, m, x) = (4, 3, 3)$ to the set (64) with $t = 2$.

If $m = 2$ or $m = 1$, then Lemma 4 gives the result $(n, m, x) = (2, 2, 1)$ or $(n, m, x) = (1, 1, 1)$, respectively. These triplets may be added, for example, to (63) with $t = 2$ and with $t = 1$, respectively.

Finally, it is easy to see that (60) has no solution with $0 < n < m$. Avoiding the repetitions we may summarise the results above as Theorem 2 states.

Proof of Theorem 3. Assume that $(n, m, x) \in \mathbb{N}^3$ with $n \geq m > 0$ is a solution of the equation

$$(65) \quad 2^n + 2^m - 1 = x^2.$$

If $m \geq 2$, then $2^n + 2^m - 1$ is a quadratic non-residue modulo 4; if $m = 1$, then apply Lemma 4 to have $(n, m, x) = (3, 1, 3)$.

Now suppose that $(n, m, x) \in \mathbb{N}^3$ is a solution of the equation

$$(66) \quad 2^n - 2^m - 1 = x^2.$$

Clearly, $n > m$ and $m < 2$. For $m = 1$ apply Lemma 4 to prove the statement.

Proof of Corollary 1. Both sequences G and H have companion polynomial $c(x) = x^2 - 3x + 2$ with zeros $x = 2$ and $x = 1$. It is well known that the terms G_m (and H_m) can be expressed in explicit form. Here by $a_G = G_1 - G_0 = 2^d + 1$ ($a_H = H_1 - H_0 = 2^d - 1$) and by $b_G = -G_1 + 2G_0 = 1$ ($b_H = -H_1 + 2H_0 = 1$) we have

$$(67) \quad G_m = a_G 2^m + b_G = 2^{m+d} + 2^m + 1,$$

$$(68) \quad H_m = a_H 2^m + b_H = 2^{m+d} - 2^m + 1.$$

Thus to determine all the squares in the recurrences G and H is equivalent to solve the equations (6) and (7) with $n = m + d$ (i.e. $n \geq m$).

Proof of Corollary 2. $\Delta_y = 4^t \Delta_x$ ($y \neq x$, $y > 0$, $x > 0$) implies

$$(69) \quad y_1^2 - 1 = 4^t (x_1^2 - 1),$$

where $y_1 = 2y + 1 \geq 3$ and $x_1 = 2x + 1 \geq 3$. In virtue of Lemma 5, (69) has no solution under the given conditions.

REFERENCES

- [1] Apéry, R., Sur une équation diophantienne, C.R. Acad. Sci. Paris Sér. A **261**, 1263-1264 (1960).
- [2] Beukers, F., On the generalized Ramanujan-Nagell equation I., Acta Arithm. **XXXVIII**, 389-410 (1981).
- [3] Cohn, E.M., Complete Diophantine solution of the Pythagorean triple $(a, b = a + 1, c)$, Fibonacci Quart., **8**, 402-405 (1970).
- [4] Gerono, C.G., Note sur la résolution en nombres entiers et positifs de l'équation $x^m = y^n + 1$, Nouv. Ann. Math. (2) **9**, 469-471 (1870); **10**, 204-206 (1871).
- [5] Hasse, H., Über eine diophantische Gleichung von Ramanujan-Nagell und ihre Verallgemeinerung, Nagoya Math. J., **27**, 77-102 (1966).
- [6] Nagell, T., The Diophantine equation $x^2 + 7 = 2^n$, Norsk. Mat Tidsskr. **30**, 62-64 (1948); Ark. f. Mat. **4**, 185-187 (1960).
- [7] Ramanujan, S., Collected papers, Cambridge Univ. Press, 327 (1927).
- [8] Shorey, T.N. - Tijdeman, R., Exponential diophantine equation, Cambridge University Press, 1986, Chapter 9., 150-168.

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